Guide to Install SQL Server on macOS Using Docker

Docker is a platform that runs applications in isolated containers, ensuring consistency across environments. Docker Desktop, its user-friendly application, simplifies container management on macOS, Windows, and Linux.

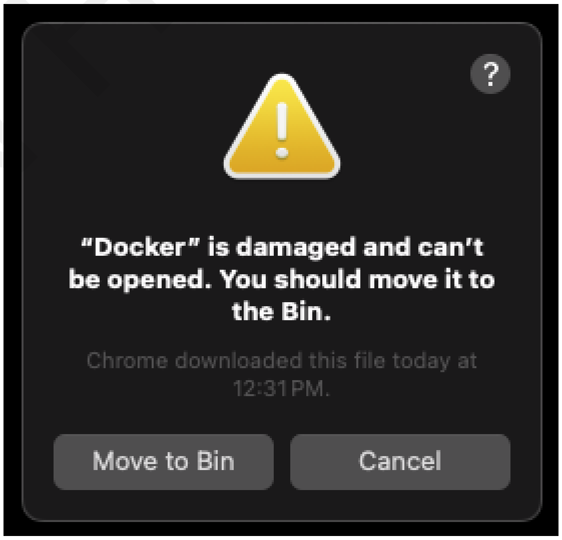
Since SQL Server is not natively supported on macOS, Docker Desktop enables you to run it in a Linux-based container. This setup ensures compatibility, ease of management, and a reproducible environment for development and learning. This guide will show you how to install SQL Server on macOS using Docker.

Step 1: Install Docker Desktop

Follow the link to download and install Docker Desktop: https://docs.docker.com/desktop/setup/install/mac-install/. Make sure that you choose the correct version that is suitable for your chip.
If you are using a Mac with an Apple silicon chip (M1/M2/M3, etc.), it is recommended that you install Rosetta 2. On your Terminal, run the following command:

```
$ softwareupdate --install-rosetta
```

If you are facing the bug that says: " "Docker" is damaged and can't be opened. You should move it to the Bin.".



First, move it to trash, then stop all the Docker processes using Activity Monitor. Then restart your computer before downloading the version 4.37.2 or later using this link.

Step 2: Download and launch the available SQL Server image.

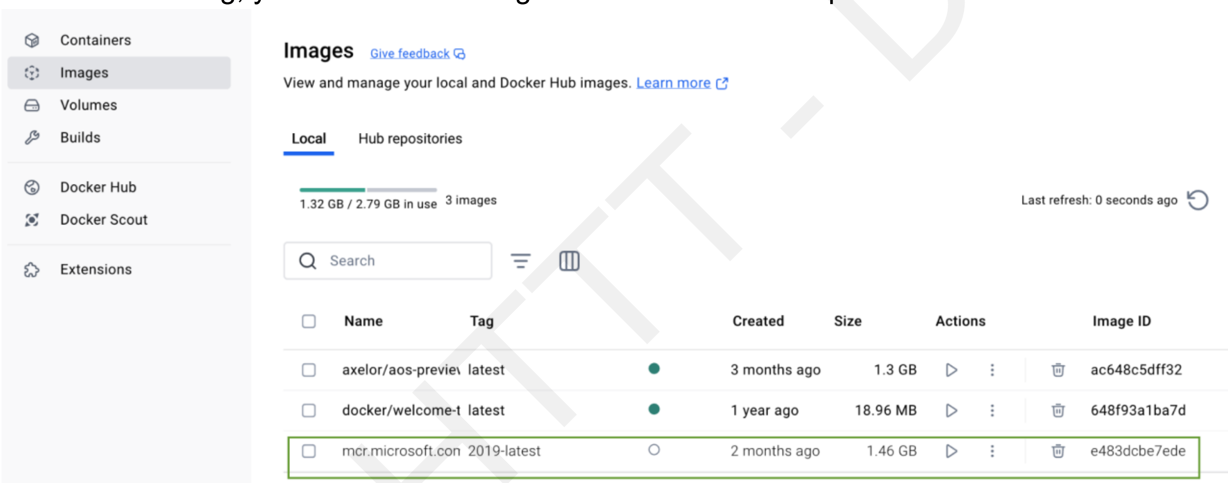
A Docker image is a prepackaged blueprint for running applications in containers. The MSSQL Docker image provides a lightweight, containerized version of Microsoft SQL Server, allowing it to run seamlessly on macOS without native installation.
https://learn.microsoft.com/en-us/sql/linux/quickstart-install-connect-docker?view=sql-server-ver15&pivots=cs1-bash&tabs=cli

Download image

On your Terminal (or Terminal inside the Docker Desktop GUI), run the following commands to pull (download) the Docker image:

```
$ docker pull mcr.microsoft.com/mssql/server:2022-latest
```

If you want to find the image for other versions of MSSQL Server, please check here.
After downloading, you can see the image in the Docker Desktop GUI.



Launch the image

Launch the image to create a docker container. A Docker container is a running instance of a Docker image. Think of the image as a recipe and the container as the prepared dish.

On the Terminal (or Docker Terminal window), run the following command:

```
$ docker run -e "ACCEPT_EULA=Y" -e "MSSQL_SA_PASSWORD=<password>" \
   -p 1433:1433 --name sql_server_test --hostname sql_server_test \
   -d \
   mcr.microsoft.com/mssql/server:2022-latest
```

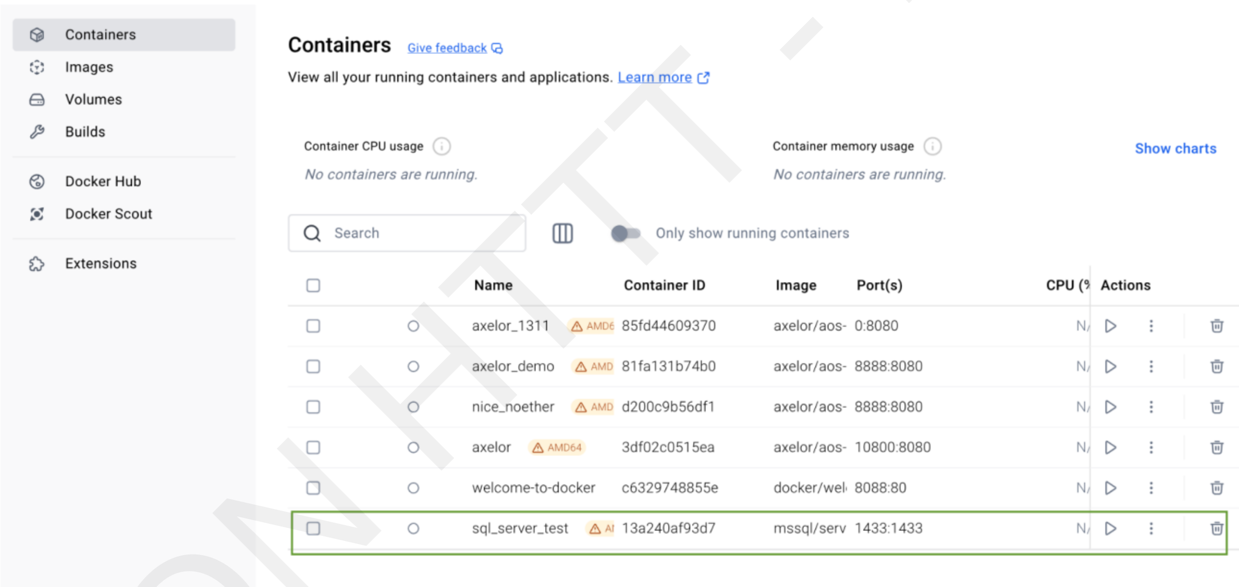
- `-e 'ACCEPT_EULA=Y'`: Confirms by agreeing with end user license agreement (EULA) for Docker.
- `-e 'SA_PASSWORD=<password>'`: Sets the database password.

- `-p 1433:1433` : Map a TCP port on the host environment (first value) with a TCP port in the container (second value). In this example, SQL Server is listening on TCP 1433 in the container and this container port is then exposed to TCP port 1433 on the host.
- `--name sql_server_test` : Sets a name for the Docker container. In this example, we are using "`sql_server_test`"
- `--hostname sql_server_test` : Used to explicitly set the container hostname.
- `-d` : Launches the docker container in daemon mode, allowing it to run in the background without a terminal window open.
- `mcr.microsoft.com/mssql/server:2022-latest` : The SQL server container image

If you're using a Mac with an ARM chip (M1, M2, etc.), you might receive the following warning:

```
"WARNING: The requested image's platform (linux/amd64) does not match
the detected host platform (linux/arm64/v8) and no specific platform
was requested
84182d3f00fb548a5cbf780221417a058b5fe61d3f599d8426eca1d65b5b0dd3"
```

However, don't worry your container will still be created after a few seconds. You can check on the Docker GUI (or using docker command: `$ docker ps -l`) for the results

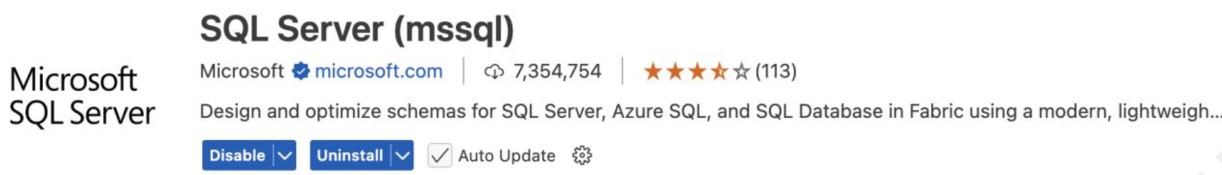


Step 3: Download and install the GUI application

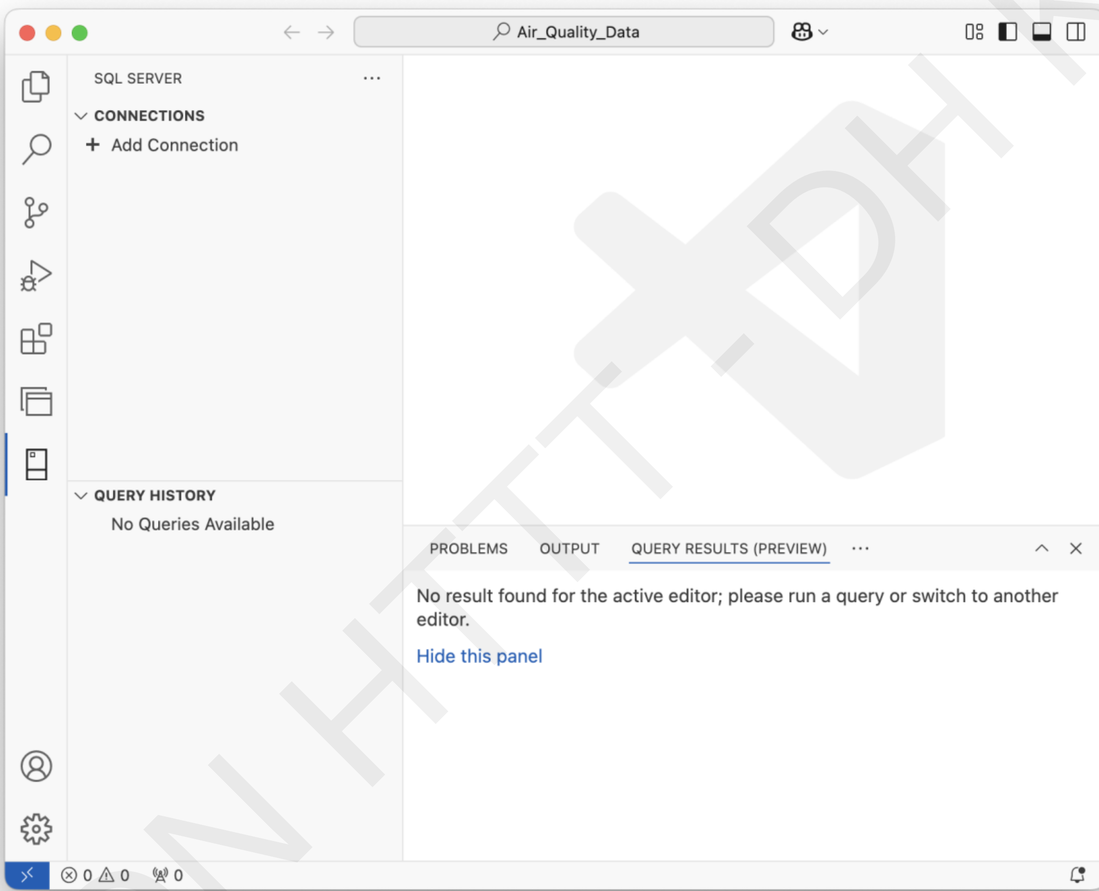
Install SQL Server (mssql) extension on Visual Studio Code

Prerequisites: Visual Studio Code

On your Visual Studio Code UI, navigate to the Extension tabs , search for sql server then install the package.



After installing the extension, you can see the SQL Server Tab at the bottom of VS Code Tab bar  , select this tab will navigate to the SQL server GUI



Create new connection to SQL Server

Next, select “+ Add Connection” to create a new connection to your SQL Server. Fill in the form with following information then select “Connect”

Profile Name: Naming your profile for this connection. For example: SQL Server 2022
Input Type: Paramaters
Server Name: localhost
Trust server certificate: True
Authentication Type: SQL login

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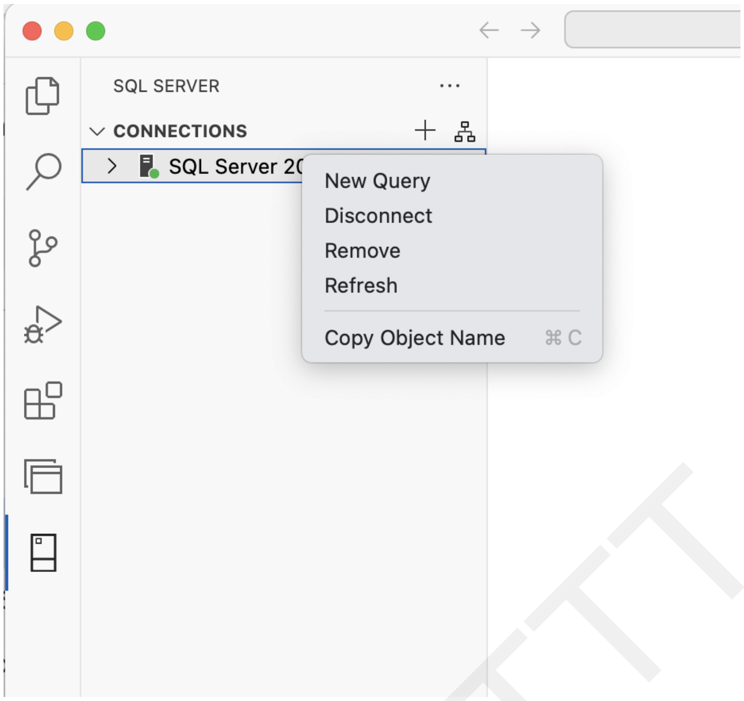
User name: sa
Password: <password> the password you specified when launching the image.

Create your first database

Right click on the Connection, in this example the “SQL Server 20222”, choose New query to create a new window to execute SQL queries.



Create new Database:

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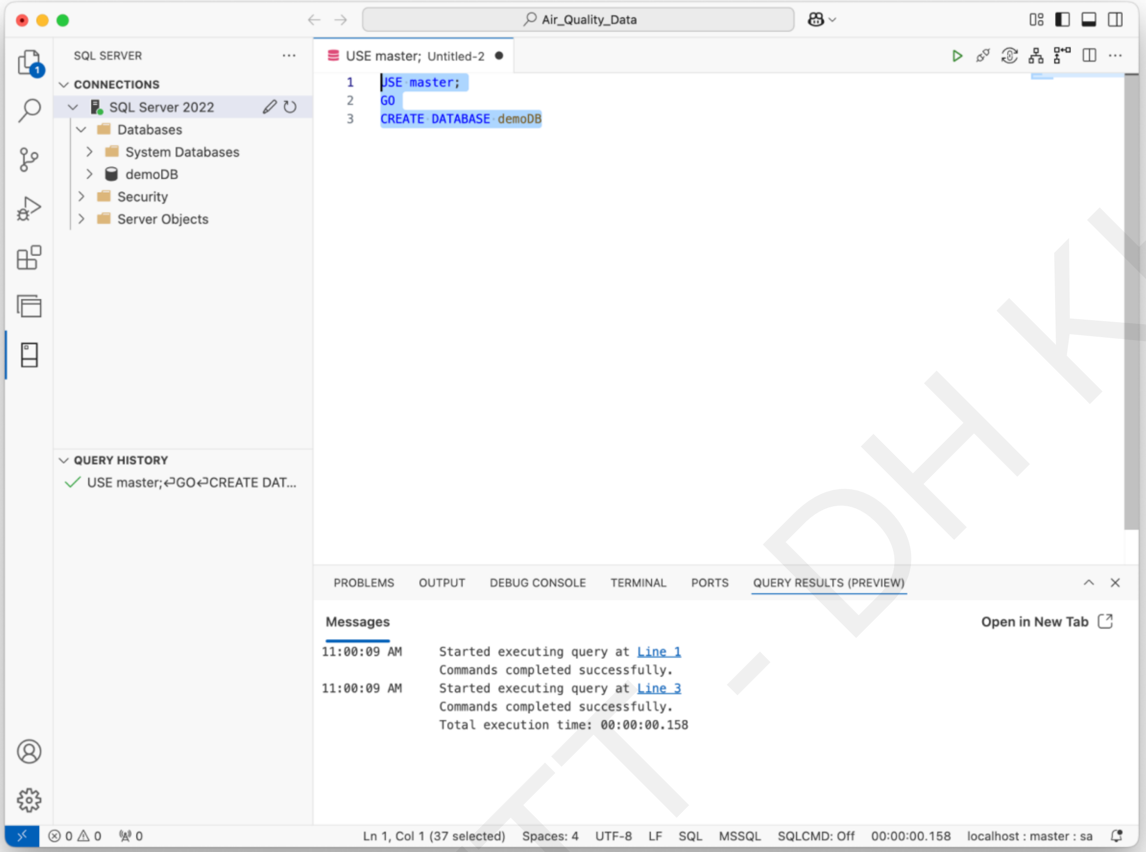

Stop and restart the SQL server

Stop working with SQL Server

When you have finished and saved your work, you can right click on the SQL Connection profile and choose Disconnect to stop the connection with the SQL Server

Then, you should stop your SQL server container as following:

- Using Docker Desktop GUI: Select the stop button of the specific container to stop that. In this example, we stop the SQL server name `sql_server_test`.

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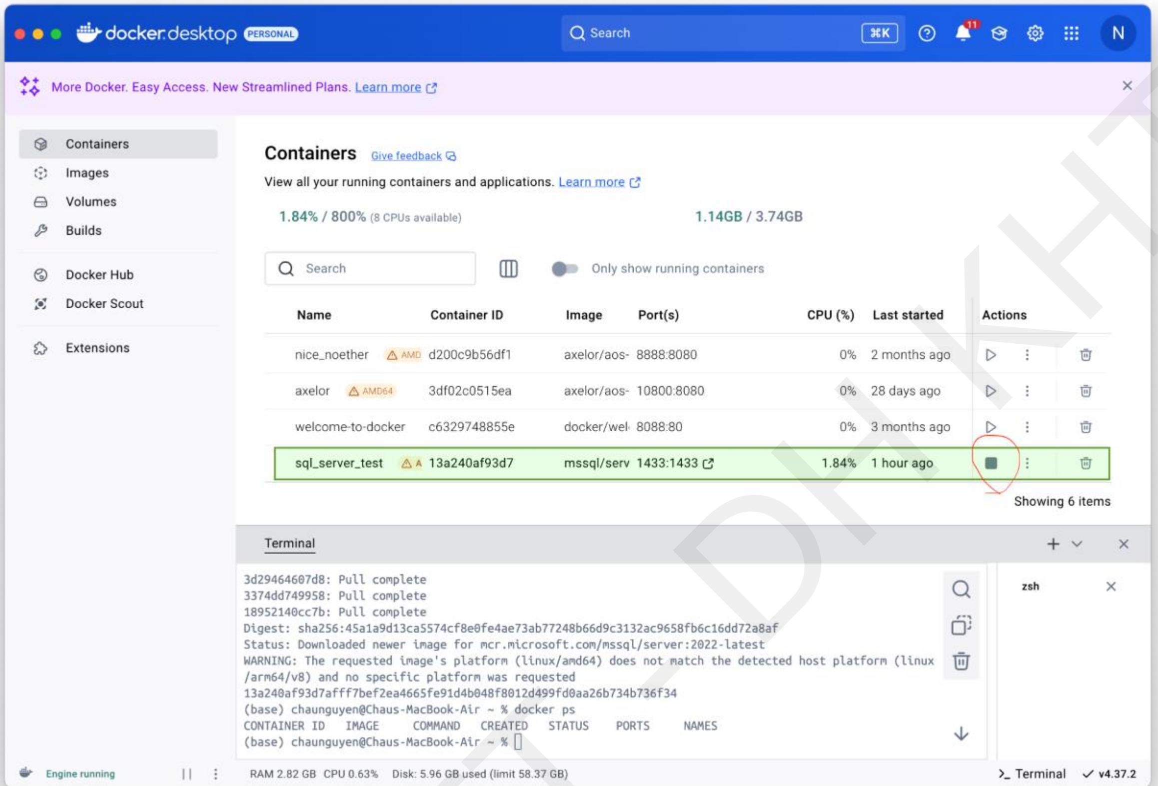

- Using Docker commands:
  `$ docker stop sql_server_test`

Restart working with SQL Server container

Restart the SQL server container using Docker GUI and reconnect the connection if you want to work again with SQL server.
Note that you only need to start the container, no need to re-launch the image.

References

1 https://learn.microsoft.com/en-us/sql/linux/quickstart-install-connect-docker?view=sql-server-ver15&pivots=cs1-bash&tabs=cli

2 https://builtin.com/software-engineering-perspectives/sql-server-management-studio-mac